Nikhil Obuleni

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MS Data Science candidate at George Washington University with 3+ years building machine learning models, deep learning systems, and ETL pipelines across humanitarian, aerospace, and NLP domains. Proficient in Python, SQL, scikit-learn, PyTorch, TensorFlow, XGBoost, PySpark, NLP, and AWS.

EDUCATION

George Washington University **Jan 2025 - Present**
M.S in Data Science | GPA: 3.75 | Award: Global Leaders (CCAS) Fellowship $17,100 (2025) *Washington, D.C.*
Coursework: Machine Learning, Data Visualization, Data Mining, Database Management, GIS, Natural Language processing.

SKILLS

Languages & ML: Python, R, SQL | scikit-learn, PyTorch, TensorFlow, XGBoost, LightGBM, ARIMA, LSTM, Random Forest, GNN, YOLOv5, Faster R-CNN, Autoencoders, Isolation Forest
Data & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Apache Spark (PySpark), Power BI, Tableau, Excel
NLP, CV & Methods: Natural Language Processing (NLP), Computer Vision, Sentiment Analysis, Topic Modeling, Statistical Modeling, Time Series Forecasting, Geospatial Analysis
Cloud & Engineering: ETL Pipelines, Apache Airflow, BigQuery, AWS (EC2, Lambda, S3), Salesforce CRM, Salesforce Flow, MongoDB, Docker, Git, REST APIs, GIS, LiDAR

WORK EXPERIENCE

GW Libraries & Academic Innovation, George Washington University **Aug 2025 - Present**
Student Data & Systems Analyst *Washington, DC*
- Manage and maintain operational records in Salesforce CRM for facility access requests and maintenance ticket lifecycles across 500+ weekly entries, ensuring data integrity and consistency across university building management systems.
- Designed custom Salesforce reports and dashboards to visualize space utilization KPIs for department leadership; automated data workflows using Salesforce Flow, reducing manual entry time by 30% and improving reporting turnaround from weekly to daily.
- Cleaned and reconciled structured data across Salesforce and university building management systems, resolving 200+ data discrepancies to ensure accurate cross-system records; provided administrative and customer service support for GW Libraries & Academic Innovation.

Data Science for Sustainable Development **Feb 2025 - Dec 2026**
Data Science Consultant (Part-time, concurrent) *Washington, DC*
- Devoloped ARIMA, LSTM, and XGBoost ensemble models on 200K+ humanitarian records spanning 18 countries, achieving 78% weighted F1 on disaster impact forecasting for field teams with limited data infrastructure.
- Standardized data sharing workflows across 6 NGO partners using Apache Airflow DAGs on BigQuery, cutting cross-organization integration time from 14 days to 48 hours and enabling near real-time reporting for humanitarian operations.
- Performed exploratory data analysis and feature engineering on multi-country humanitarian datasets; designed evaluation metrics aligned with field deployment constraints, ensuring model outputs were interpretable and actionable for non-technical stakeholders.

Asteria Aerospace **May 2023 - Aug 2023**
Data Science Intern *Bangalore, India*
- Generated computer vision models using YOLOv5 and Faster R-CNN for real-time drone object detection, enhancing surveillance efficiency by 30%, improving target recognition accuracy across varied environmental conditions.
- Deployed predictive maintenance models using LSTMs and Random Forest to forecast drone failures, reducing unexpected breakdowns by 25%; devised an anomaly detection system using Autoencoders and Isolation Forest, improving fault detection by 40%.
- Conducted geospatial data analysis using GIS, LiDAR, and satellite imagery, improving drone navigation accuracy by 20% and reducing collision risks by 15%; communicated model results to engineering and operations stakeholders.

Blackoffer **Sep 2022 - Nov 2022**
Data Science Intern *Remote*
- Applied advanced NLP techniques including sentiment analysis, keyword extraction, and topic modeling on 50,000+ text records using Python, deriving actionable insights that improved client campaign targeting accuracy by 18% and enabled more effective audience segmentation strategies.
- Extracted and processed large-scale textual data from diverse online sources using Python-based web scraping tools (BeautifulSoup, Scrapy), fully automating data collection pipelines that previously required significant manual effort, reducing processing time by 40%.
- Accelerated machine learning models for text classification and clustering; optimized ETL data pipelines to ensure data quality and analytical efficiency across multiple client projects.

PROJECTS

Retail Demand Forecasting & Price Optimization
- Engineered end-to-end demand forecasting system using Python, PySpark, and XGBoost achieving 93% accuracy and 35% stockout reduction across a multi-store retail chain; A/B-tested a markdown optimization algorithm improving profitability by 12%.

Netflix - Content Recommendation & Churn Prediction
- Created a GNN and Collaborative Filtering recommendation engine on 10M+ viewer records boosting engagement by 35%; built XGBoost and LSTM/ANN churn prediction model achieving 92% accuracy, enabling data-driven retention strategies that reduced subscription churn by 20%.

Real-Time Traffic Accident Prediction
- Streamlined LSTM and Prophet time-series models to predict congestion and accident frequency across city regions; performed geospatial analysis using GIS, Folium, and Plotly Maps to identify high-risk accident hotspots from real-time public datasets.